

Chapter 7 Homework Answer Sheet

Part A Multiple choice (15 points)

1-5 ACADD

6-10 EBCAD

11-15 CECBA

Part B Short questions (20 points)

16. (3 points)

(a) (1 pt) Cesium **or** Cs

(b) (2 pts) $2 \text{Cs}(s) + \text{Cl}_2(g) \rightarrow 2 \text{CsCl}(s)$

17. (4 points)

(a) (2 pts) Cl, S, K

(b) (2 pts) K^+ , Cl^- , S^{2-}

18. (4 points)

(a) (2 pts) $Z_{\text{eff}} = Z - S = 16 - 10 = 6+$

(b) (2 pts) $Z_{\text{eff}} = Z - S = 16 - (1.00 \times 2 + 0.85 \times 8 + 0.35 \times 5) = 5.45+$

19. (5 points)

(a) (2 pts) First ionization energy of F^- : $\text{F}^-(g) \rightarrow \text{F}(g) + e^-$

Electron affinity of F: $\text{F}(g) + e^- \rightarrow \text{F}^-(g)$

(b) (2 pts) The I_1 of F^- is positive, whereas EA of F is negative.

(c) (1 pt) Yes, these two quantities are equal in magnitude.

20. (4 points)

(a) (1 pt) The first and second ionization energies of calcium are less than those of beryllium.

(b) (3 pts) The electron configuration of C is $1s^2 2s^2 2p^2$. When added an electron, it becomes more stable with half-filled p subshells. However, the N atom has half-filled p subshells. The added electron must be put in an orbital that is already occupied, resulting in larger electron-electron repulsions. Consequently, the C atom has more negative electron affinity than N atom.